AP CALCULUS AB

Unit 1 • Worksheet 1-8A • Answer key

Worksheet 1-8A • MVT

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What does the MVT need on $[a, b]$?

(A) f continuous on $[a, b]$ and differentiable on (a, b)

(B) f increasing

(C) $f(a) = f(b)$

(D) f continuous on (a, b) only

Solution

Answer: (A). Both conditions.

Question 2

NO CALCULATOR

f is differentiable, $f(0) = -2$ and $f(4) = 10$. The MVT guarantees a c in $(0, 4)$ with $f'(c) =$

(A) 12

(B) 3

(C) 2

(D) $\frac{1}{3}$

Solution

Answer: (B). $\frac{10-(-2)}{4} = 3$.

Question 3

NO CALCULATOR

For which function does the MVT apply on $[0, 2]$?

(A) $|x - 1|$

(B) $\frac{1}{x-1}$

(C) $x^2 - 3x$

(D) $\begin{cases} x, & x < 1 \\ x + 1, & x \geq 1 \end{cases}$

Solution

Answer: (C). A polynomial. The others have a corner, an asymptote or a jump at $x = 1$.

Question 4

NO CALCULATOR

h is differentiable.

x	1	4	6
$h(x)$	2	11	5

The MVT guarantees a c in $(4,6)$ with $h'(c) =$

(A) 3
(B) 6
(C) -6
(D) -3

Solution
Answer: (D). $\frac{5-11}{6-4} = -3$.

Question 5

NO CALCULATOR

f is differentiable and $f(1) = f(5) = 3$. What does the MVT guarantee?

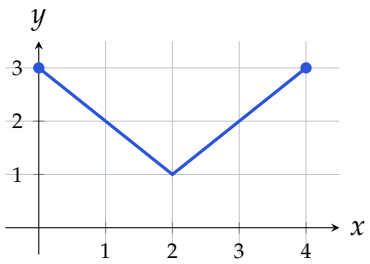
(A) A c in $(1,5)$ with $f'(c) = 0$ (a horizontal tangent)
(B) $f(3) = 3$
(C) f is constant.
(D) A c in $(1,5)$ with $f(c) = 0$

Solution
Answer: (A). Average rate $\frac{3-3}{4} = 0$.

Question 6

NO CALCULATOR

The graph of $g(x) = |x-2| + 1$ on $[0,4]$ is shown. Does the MVT guarantee a c in $(0,4)$ with $g'(c) = \frac{g(4)-g(0)}{4-0}$?



(A) Yes, $c = 2$.
(B) No: g is not differentiable at $x = 2$.
(C) Yes, because g is continuous.
(D) No, because $g(0) = g(4)$.

Solution
Answer: (B). The corner at $x = 2$ breaks the conditions; the slopes are only -1 and 1, never 0.

Question 7

NO CALCULATOR

In words, the MVT says that somewhere the slope of a _____ equals the slope of the _____.

(A) secant line; tangent line
(B) tangent line; x-axis
(C) tangent line; secant line
(D) graph; table

Solution
Answer: (C). Instantaneous rate (tangent) = average rate (secant).

Question 8 (Free response)

NO CALCULATOR

The water level $W(t)$, in meters, is differentiable.

t (h)	0	4	6	10
$W(t)$	2	5	9	7

(a) Explain why there must be a time t in $(4,6)$ with $W'(t) = 2$.
(b) Must there be a time in $(0,10)$ when the level rises at 0.5 m/h? Justify.
(c) Does the MVT apply to $g(x) = |x-3|+1$ on $[1,5]$? Explain.

Solution
(a) W is differentiable (so continuous) on $[4,6]$; $\frac{9-5}{2} = 2$: MVT.
(b) Yes: $\frac{7-2}{10} = 0.5$, MVT on $[0,10]$.
(c) No: g has a sharp corner at $x = 3$, so it is not differentiable on $(1,5)$.